

URYUPINO, Russian Federation

WMO: 307810

Lat: 52.750N

Lon: 120.033E

Elev: 370

StdP: 96.96

Time Zone: 9.00 (E09)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.4	(c) -39.6	(d) -43.6	(e) 0.1	(f) -41.2	(g) -41.6	(h) 0.1	(i) -39.3	(j) 3.3	(k) -17.6	(l) 2.4	(m) -19.4	(n) 0.1	(o) 270	(p) 0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	30.4	19.5	28.2	18.7	26.2	17.9	21.4	26.8	20.3	25.4	19.3	24.1	1.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	15.1	24.0	18.6	14.1	22.5	17.6	13.2	21.3	64.5	26.6	60.3	25.4	56.8	24.2	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.8	3.9	3.2	-43.9	34.2	2.7	1.9	-45.8	35.5	-47.3	36.6	-48.8	37.7	-50.8	39.0
			-44.0	23.9	2.5	2.0	-45.8	25.3	-47.2	26.5	-48.6	27.6	-50.4	29.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.3	-28.3	-21.9	-11.5	0.3	8.8	15.8	18.6	16.2	8.5	-2.3	-17.4	-27.9
	DBStd	17.47	4.45	5.32	6.83	4.60	3.76	3.48	2.85	3.48	4.29	5.43	7.38	5.05
	HDD10.0	5571	1187	893	665	292	68	3	0	3	78	382	823	1176
	HDD18.3	7979	1446	1126	923	542	297	91	31	81	296	641	1073	1434
	CDD10.0	703	0	0	0	1	31	176	267	197	33	0	0	0
	CDD18.3	70	0	0	0	0	1	13	40	16	0	0	0	0
	CDH23.3	1222	0	0	0	1	53	398	508	249	14	0	0	0
	CDH26.7	362	0	0	0	0	10	124	161	66	1	0	0	0
Wind	WSAvg	0.8	0.2	0.3	0.9	1.3	1.4	0.9	0.7	0.7	0.9	0.8	0.5	0.2
Precipitation	PrecAvg	428	5	5	3	17	37	66	108	100	53	16	9	8
	PrecMax	576	15	25	14	54	91	180	213	219	111	67	17	27
	PrecMin	308	1	0	0	1	3	19	25	38	9	1	2	1
	PrecStd	79	3	6	4	14	24	35	45	44	26	15	4	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.8	0.0	10.5	20.8	27.7	33.0	32.9	31.3	25.2	16.5	2.6	-10.3
		MCWB	-14.0	-4.0	2.8	9.1	14.3	19.1	20.2	20.4	15.8	8.2	-0.9	-11.3
	2%	DB	-15.5	-4.1	6.9	16.5	24.6	30.1	30.5	28.6	22.4	12.7	-0.6	-14.7
		MCWB	-16.3	-6.7	1.0	6.6	12.3	18.6	20.2	19.8	14.3	5.6	-2.9	-15.3
	5%	DB	-18.1	-8.3	3.6	13.2	21.7	27.6	28.3	26.1	19.8	10.0	-3.1	-17.4
		MCWB	-18.7	-10.0	-1.1	4.7	11.0	17.4	19.6	18.7	13.3	4.2	-4.8	-17.8
	10%	DB	-20.9	-11.5	0.9	10.3	19.0	25.4	26.2	23.9	17.4	7.2	-5.7	-19.8
		MCWB	-21.2	-12.7	-2.7	3.1	9.9	16.7	19.2	17.7	11.9	2.8	-7.2	-20.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-13.8	-3.8	3.2	9.2	15.6	22.1	23.1	22.4	17.9	8.4	-0.3	-10.8
		MCDB	-12.8	-0.3	9.3	19.3	23.8	28.8	28.7	27.9	22.2	15.6	1.8	-10.2
	2%	WB	-16.3	-6.7	1.1	6.8	13.5	20.0	21.9	20.8	16.0	6.2	-2.9	-15.3
		MCDB	-15.5	-4.2	6.8	15.3	21.8	26.4	27.3	26.1	19.9	11.6	-1.0	-14.8
	5%	WB	-18.7	-10.0	-0.8	5.1	12.0	18.7	20.8	19.7	14.5	4.6	-4.8	-17.8
		MCDB	-18.1	-8.4	3.3	12.1	20.2	25.4	25.7	24.5	18.0	9.4	-3.1	-17.4
	10%	WB	-21.2	-12.7	-2.5	3.5	10.5	17.7	19.8	18.5	12.9	3.0	-7.2	-20.1
		MCDB	-20.8	-11.5	0.8	9.4	17.8	24.0	24.4	22.4	16.6	7.4	-5.8	-19.8
Mean Daily Temperature Range	5% DB	MDBR	14.7	19.9	20.3	16.8	17.3	17.4	14.4	13.9	15.8	15.6	14.9	13.4
		MCDBR	15.7	21.8	20.8	22.6	23.8	22.3	19.4	18.5	19.6	19.1	15.6	14.0
		MCWBR	14.8	19.1	15.0	13.2	12.3	11.2	9.7	10.1	11.9	12.2	13.0	13.3
	5% WB	MCDBR	15.6	21.5	19.7	20.1	20.2	18.6	15.7	15.3	15.5	18.6	15.2	14.0
		MCWBR	14.7	18.8	14.3	12.1	10.9	10.2	8.8	9.1	10.2	12.0	12.8	13.3
Clear-Sky Solar Irradiance	taub	0.261	0.290	0.340	0.463	0.535	0.500	0.452	0.423	0.369	0.333	0.290	0.271	
	taud	2.152	2.118	2.054	1.892	1.801	2.005	2.212	2.269	2.355	2.325	2.214	2.085	
	Ebn at Noon	718	806	831	761	725	761	796	799	805	754	672	610	
	Edh at Noon	68	97	129	174	201	166	133	119	96	80	65	61	
All-Sky Solar Radiation	RadAvg	1.35	2.48	4.10	5.38	5.37	5.90	5.32	4.65	3.56	2.39	1.48	1.03	
	RadStd	0.04	0.06	0.15	0.30	0.41	0.49	0.45	0.41	0.29	0.16	0.06	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page